F1 Driver Performance Under Adverse Conditions

Does Skill or Car Matter More?

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Why This Matters

The Problem

In dry conditions on permanent circuits, the car is the overwhelming performance determinant.

This makes it nearly impossible to separate genuine driver talent from machinery advantage using raw results alone.

The Hypothesis

Adverse conditions (rain, street circuits, windiness, high altitude) act as an “equalizer” on car performance, and drivers’ true skills are able to shine through.

Target Audience: F1 team principals, talent scouts, and data journalists evaluating driver value

Research Questions

1. Which drivers consistently perform better in wet conditions than dry races?
2. Which drivers perform better in street circuits than permanent tracks?
3. Which drivers consistently display the ability to outperform their teammates?
4. Can we predict whether a driver will beat their teammate from pre-race conditions alone?
5. How much of finishing position is explained by the car vs. the driver?

Data Sources

Dataset 1: Race Results & Qualifying

Source: Jolpica F1 API

Finishing positions, grid slots, statuses
2022-2025 seasons (400 driver-race rows)

Dataset 2: Session Weather

Source: OpenF1 API

Rainfall, track temp, humidity per session
Classified: Wet (>30%), Mixed (5-30%), Dry (<5%)

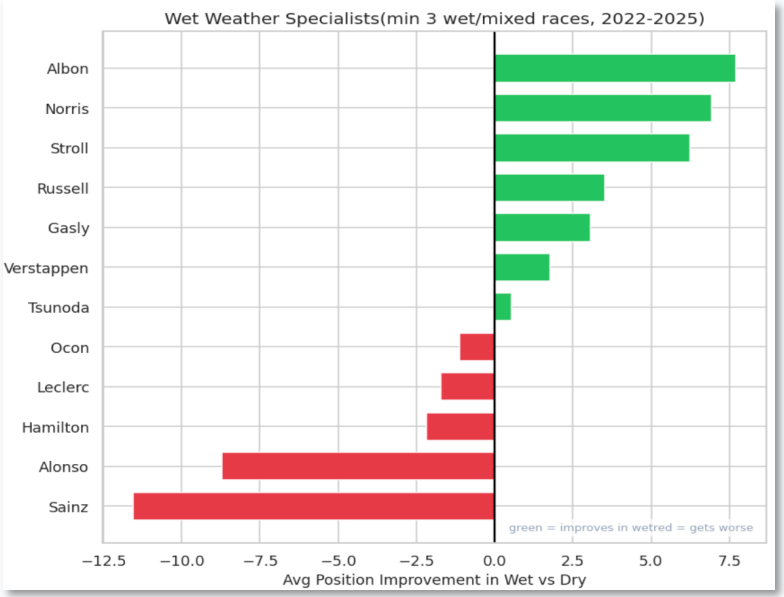
EDA – Drivers Dominating Wet Races

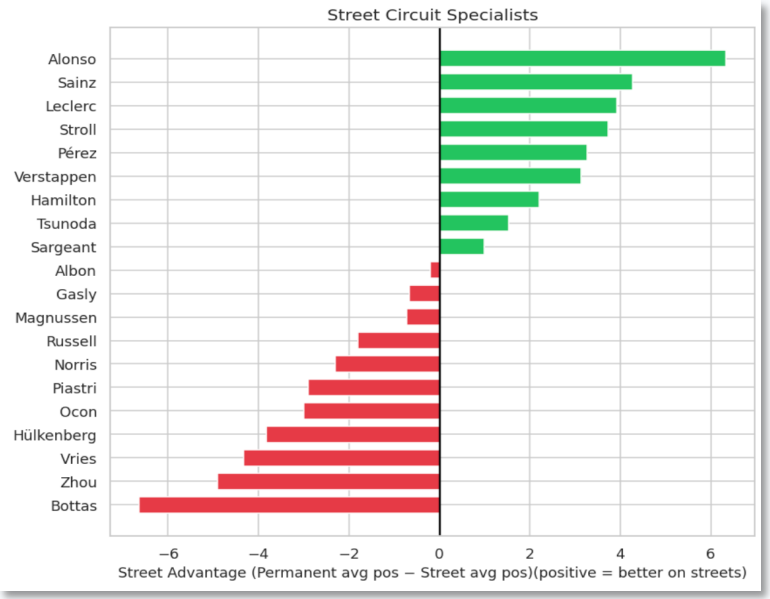
Methods

.groupby + .agg
Calculated differences in average points, position finished between dry and wet condition to measure improvement

Key Finding

Street and wet specialists partially overlap but are distinct groups. Perez (+3.3) and Leclerc (+3.9) show strong street advantages along with high beat-teammate rates, confirming genuine skill. Qualifying performance (qual_pos) correlates more strongly with results on street circuits than on permanent tracks.





DNF Rates & High-Risk Circuits

Highest DNF Circuits

Albert Park	25.3%
Shanghai	17.5%
Jeddah	16.2%
Miami	12.2%
Bahrain	10.0%

Best Performers at High-DNF Circuits

Norris, Piastri, Verstappen, and Russell perform best at high-DNF circuits.

EDA – Teammate Gap Analysis

Using beat TM rate, we the variable race car is controlled, and we can test for driver skill

Driver	Team Tier	Beat TM %	Avg Pos Gain	Races
George Russell	Top	80%	+0.1	20
Alexander Albon	Mid	79%	+1.2	19
Max Verstappen	Top	75%	-1.1	20
Yuki Tsunoda	Mid	71%	0.0	17
Lando Norris	Top	65%	+0.1	20
Charles Leclerc	Top	65%	-2.0	20
Lewis Hamilton	Top	20%	+0.5	20
Lance Stroll	Mid	30%	+1.4	20

Insight

Russell (80%) and Albon (79%) consistently dominate their teammates. Hamilton's 20% rate reflects a difficult period against Russell at Mercedes.

Model 1: Predicting Podium Finishes

Logistic Regression: Can we predict P1-P3 from pre-race conditions?

88.75%
Accuracy

0.934
ROC-AUC

58%
Podium Recall

Feature	Coefficient	Odds Ratio	Interpretation
log_grid	**-1.61**	0.19	Starting further back sharply reduces podium probability
tier_enc	**-1.41**	0.24	Worse car tier reduces podium odds
condition_enc	**+0.30**	1.36	Wet conditions increase podium odds (upsets more likely)

Model 2: Predicting Finishing Position

R-squared: **0.365**

MAE: 3.67 positions

~63% unexplained

Driver skill, strategy, incidents, luck

Why?

Key Findings

Wet weather reveals hidden talent

Drivers like Albon (+7.7) and Norris (+6.9) consistently overperform in rain relative to their dry baseline, regardless of car tier.

Street specialists are a distinct group

Street circuits reward qualifying and precision. Perez, Leclerc, and Alonso show persistent street advantages after controlling for constructor tier.

Adverse conditions amplify skill gaps

Beat-teammate advantages grow in wet conditions. Back-marker teams are disproportionately harmed (wet_x_tier interaction), suggesting fewer wet-weather development resources.

Limitations & Future Work

Limitations

- Small wet race sample
- Driver-constructor pairings
- Weather aggregated at session level: intra-race changes (e.g. rain starting lap 30) not captured
- Safety car periods

Future Work

- Sector-level pace data (OpenF1 car_data) for speed-based performance measurement
- Expand dataset
- Tyre compound data to separate strategy-driven position gains from on-track skill

Thank You